Determinants of Vehicle (Car)Valuation in the Sri Lankan Online Secondary Market

A Multi-Dimensional Analysis of the Pre and Early-2025 Import Liberalization Phases

This study analyzes how depreciation, performance, utility, and market sentiment influenced used car prices in Sri Lanka during the pre- and early-2025 import liberalization phases, capturing both speculative pricing and early market corrections.

Group 07

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Abstract

This study investigates the determinants of used vehicle prices in Sri Lanka during the pre- and early-2025 import liberalization period, a phase marked by extreme market volatility. Using a dataset of 9,676 online listings, a Generalized Additive Model (GAM) captures the complex, non-linear relationships that rendered traditional OLS regression unsuitable. Results show that brand equity, vehicle age, and engine capacity are the primary drivers of valuation, with Toyota serving as a mid-market benchmark for value retention while budget brands like Tata experience the steepest depreciation. Mechanical features, including hybrid and electric powertrains, command significant premiums, and standardized comfort features continue to act as price multipliers. Geo-economic analysis indicates a nationally integrated market, with no statistically meaningful differences between urban and non-urban locations. These findings provide a robust, data-driven framework for establishing fair listing prices and mitigating information asymmetry in a highly speculative secondary vehicle market.

Introduction

Sri Lanka's vehicle market exhibited abnormal pricing behavior during the 2020–2024 import ban, where prolonged supply restrictions and inflation caused used vehicles to appreciate rather than depreciate. The announcement of import liberalization in early 2025 shifted the market into a speculative correction phase, and in the absence of an official price guide, pricing became inconsistent, creating a widening gap between seller expectations and buyer valuations.

This study examines a cross-sectional dataset of vehicle listings from December 3, 2024, to February 5, 2025, capturing pre-liberalization market volatility. To date, no reliable statistical model exists to estimate fair listing prices in Sri Lanka's digital used vehicle market, where conventional depreciation models fail to reflect consumer preferences for brand reliability and resale value. This study addresses that gap by analyzing prices during the pre-liberalization period across four key drivers: depreciation, mechanical performance, comfort features, and market dynamics, revealing the factors that truly determine value.

About the data set

This dataset contains web scraped car listings from **ikman.lk** and **Riyasewana.com**, two of the most popular online vehicle marketplaces in Sri Lanka.The dataset contains 9,788 records and 16 variables, of which 12 are categorical (brand, model, fuel type, transmission gear, etc.), while 4 are numerical, including year of manufacture, engine capacity (cc), mileage (km), and vehicle price.

Dataset link:
https://www.kaggle.com/datasets/prasadnirmal/srilankan-second-vehiclecar-price-dataset

Variable	Definition	Data Type	Details / Range
Brand	The manufacturer of the vehicle	Categorical	50 vehicle brands
Model	The specific model of the car	Categorical	1,555 Models
YOM	Year of Manufacture	Numerical	Discrete (e.g., 2022)
Engine (cc)	Engine size in cubic centimeters	Numerical	Continuous
Gear	Type of transmission	Categorical	Automatic or Manual
Fuel Type	Type of fuel used	Categorical	Diesel, Petrol, Electric, Hybrid
Mileage (KM)	Total distance traveled	Numerical	Continuous
Town	Sale location	Categorical	107 towns
Date	Listing post date	Date	YYYY-MM-DD
Leasing	Leasing options availability	Categorical	Available / Not Available
Condition	Overall status of the car	Categorical	Used or New
Air Condition	Presence of AC	Categorical	Available / Not Available
Power Steering	Presence of power steering	Categorical	Available / Not Available
Power Mirror	Presence of power mirrors	Categorical	Available / Not Available
Power Window	Presence of power windows	Categorical	Available / Not Available
Price	Selling price in lakhs (LKR)	Numerical	Continuous

```
--- Dataframe info ---
<class 'pandas.core.frame.DataFrame'>
Index: 9788 entries, 0 to 9788
Data columns (total 16 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Brand          9788 non-null   object
 1   Model          9788 non-null   object
 2   YOM            9788 non-null   int64
 3   Engine (cc)    9788 non-null   float64
 4   Gear           9788 non-null   object
 5   Fuel Type      9788 non-null   object
 6   Millage(KM)    9788 non-null   float64
 7   Town           9788 non-null   object
 8   Date           9788 non-null   object
 9   Leasing        9788 non-null   object
 10  Condition      9788 non-null   object
 11  AIR CONDITION  9788 non-null   object
 12  POWER STEERING 9788 non-null   object
 13  POWER MIRROR   9788 non-null   object
 14  POWER WINDOW   9788 non-null   object
 15  Price          9788 non-null   float64
dtypes: float64(3), int64(1), object(12)
memory usage: 1.3+ MB
None

--- Number of rows and columns ---
Rows: 9788, Columns: 16
```

Figure 1: Data Description

Problem definition and objectives of the study

This study aims to identify and quantify the factors associated with second-hand vehicle prices in Sri Lanka during the closed market era (2020–2024), helping stakeholders make informed pricing decisions amid high information asymmetry. Without a standardized valuation system, prices vary widely even for identical vehicles. This project addresses the lack of a reliable "Fair Listing Price" model by focusing on four key drivers as follows:

1. **Analyze Brand Equity and Depreciation to assess its impact on pricing.**
2. **Evaluate Mechanical Performance features and their influence on value.**
3. **Examine Comfort Features and their effect on vehicle pricing.**
4. **Study Market Sentiment and Geo-Economic factors to understand price variability.**

Data preprocessing

```
Number of duplicates: 18
Number of missing values: 0
Dataset shape after cleaning: (9770, 16)
```

Figure 2 : Data preprocessing

Before conducting the EDA and advanced analysis, the dataset was carefully checked and cleaned for quality. 18 duplicate rows were detected and removed to ensure data integrity.There were no missing values across any of the 16 variables.After cleaning the dataset comprises 9,770 records and 16 attributes. All variables were checked for appropriate data types and no changes were required. Summary statistics of all variables were reviewed to identify unrealistic or inconsistent values but none were detected.

The 'Date' variable, indicating when a vehicle was posted, was removed as it provided no predictive information for the 'price' target, simplifying the feature set. Under the "Condition" variable, vehicles labeled as "New" were excluded from the dataset, as this project focuses on the used car market. After completing all preprocessing steps,the final dataset contains 9,676 observations and 15 variables.

Feature Engineering

1. **Age**

Vehicle age was derived from 'YOM' (year of manufacture) by subtracting 'YOM' from the reference year(2026).

Age = Reference Year (2026) − Year of Manufacture (YOM)

2. **Year Category**
The year of manufacture(YOM) variable was converted into a categorical variable called **Year_Category** to classify vehicles based on their year of manufacture.Vehicles were grouped into 3 categories,
    - Old: YOM is before 2010 (YOM < 2010)
    - Intermediate :YOM is from 2010 to 2017 (2010 ≤ YOM ≤ 2017)
    - Modern :YOM is 2018 or later (YOM ≥ 2018)
3. **Engine Segment**
A new categorical variable "**Engine_Segment**" was created by using the Engine (cc) variable.
    - Micro (< 800 cc)
    - Compact (800-1200 cc)
    - Mid-Range (1200-1600 cc)
    - Large (>1600 cc)

4. **Converted the leasing column from text to a boolean format**

The leasing variable originally contained two categories such as "No leasing" and "ongoing leasing".This variable was converted to boolean representation.

    - Entries containing "Ongoing leasing" were labeled True
    - Entries containing "No leasing" were labeled False
5. **Province-Level Feature Engineering**
The original 'town' variable with 107 levels was mapped to provinces to reduce high cardinality, creating a new **'Province'** variable. Unmatched towns were grouped under 'Other.'
6. **Urban Non urban Location classification**
A new categorical variable, **'Location Type,'** was created from 'town,' labeling towns under municipal or urban councils as 'Urban' and all others as 'Non Urban.'
7. **Brand–Model Feature Engineering**
1) **Create 'brand_grouped' using 'brand'**
The dataset contains many vehicle brands, several appearing rarely. Therefore, brands were grouped based on their frequency of occurrence using a defined method.
    - Brands with >100 listings retained as individual categories.
    - Brands with 10–100 listings grouped as "OTHER".
    - Brands with <10 listings grouped as "RARE".
2) **Composite Brand–Model Variable ,create 'simple model' and 'brand model' variable**
"A 'simple_model' variable was created from the first word of the original model name and combined with 'brand_grouped' to form a 'brand_model' feature, standardizing names, reducing sparsity, and preserving key brand-model distinctions.

Basic Exploratory Data analysis Results

Investigate Response Variable
The distribution of vehicle prices is highly right-skewed, with most listings in the lower range and a long tail of high-value vehicles. This asymmetric, heavy-tailed distribution indicates deviation from normality, highlighting the presence of few expensive vehicles amid many affordable ones.

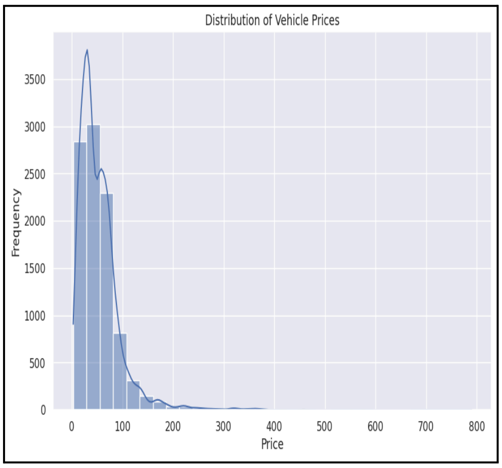


Figure 3: Distribution of Price

Furthermore, the distribution of the price data was analyzed for bimodality, which might suggest the existence of multiple market segments. To examine bimodality, three complementary statistical approaches, the Bimodal Coefficient, Ashman's D value and Hartigan's Dip Test were employed.

The Bimodal Coefficient was 0.409, below the 0.555 threshold, indicating the price distribution lacks a clear or pronounced bimodal shape. The Ashman's D value of 1.381 shows poor separation between distributions, suggesting overlapping modes and weak bimodality. The Hartigan's Dip Test gave a Dip of 0.014 with p-value 0.000, rejecting unimodality. Together, these results indicate the distribution is not strongly bimodal.

Distributional Characteristics and Categorical Structures of the Data

Numerical variables

The numerical variables exhibit substantial skewness and the presence of extreme values, reflecting the heterogeneous nature of the vehicle market.

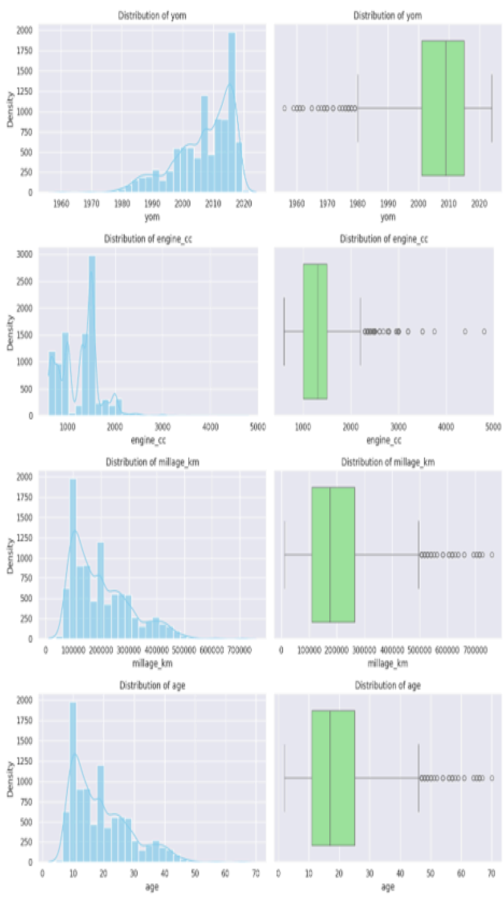


Figure 4 : Distribution of numerical variables

The distribution of **Year of Manufacture (YOM)** indicates a high concentration of vehicles manufactured in recent years, while older models appear as scattered data points**.**

Engine capacity displayed a right skewed distribution with most vehicles clustered in the lower engine segments and a small number of high-capacity vehicles forming a long tail. **Mileage** and **vehicle age** variables also exhibit pronounced right skewness, with most vehicles concentrated at lower mileage and age levels, alongside a gradual decline toward higher values. The corresponding boxplots highlight several extreme observations, indicating the presence of heavily used or very old vehicles within the market.

Feature	Skewness
price	3.407278
age	1.053616
millage_km	1.053616
engine_cc	0.497671
price_transformed	-0.234606
log_engine_cc	-0.371510
yom	-1.053616

The skewness levels depicted in this figure are in accordance with the levels depicted in the distribution plots of each numerical variable. For example, variables like price, mileage, and age depict high levels of positive skewness, which confirms the right-skewed distribution and high-value outlying observations depicted in the earlier histograms and boxplots.

Categorical variables

The **grouped brand distribution** is highly concentrated, dominated by Toyota and Suzuki,followed by Nissan, with all other brands representing much smaller shares.The **fuel type distribution** shows a majority of petrol vehicles, a notable share of hybrids, and some diesel, while electric vehicles remain negligible, reflecting growing adoption of fuel efficient technologies. Most vehicles have automatic as **gear type**, indicating a preference for convenience. Additionally, 95.8% are not leased, power steering is nearly universal (91.2%), and power mirrors are present in 74.5% of cars.

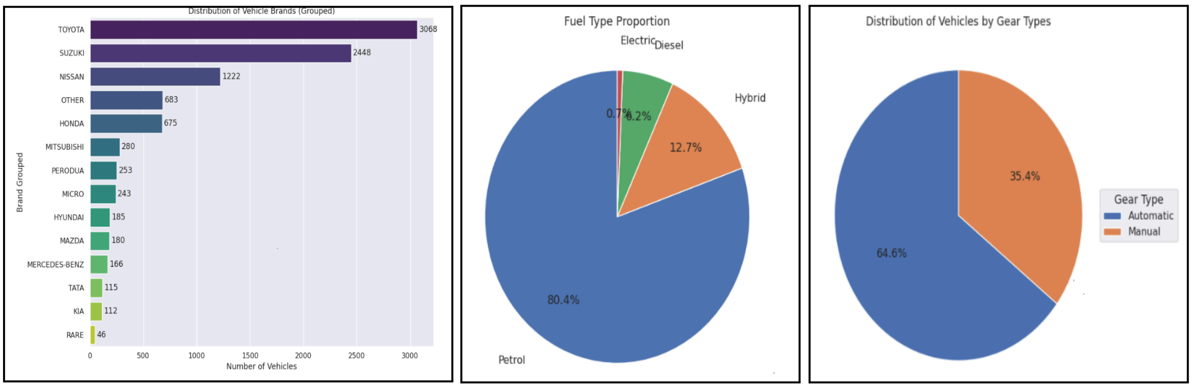


Figure 5 :Distribution of vehicles by brand, fuel type, and gear type.

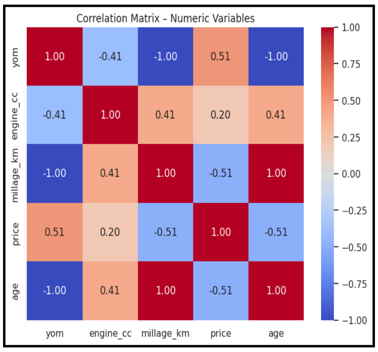
Outlier Identification and Handling

Feature	Q1	Q3	IQR	Lower_Bound	Upper_Bound	Outlier_Count	Outlier_Percentage
price	26.3375	69.7625	43.425	-38.8	134.9	381	3.937578
engine_cc	1000.0000	1500.0000	500.000	250.0	2250.0	109	1.126499
yom	2001.0000	2015.0000	14.000	1980.0	2036.0	75	0.775114
millage_km	110000.0000	264000.0000	154000.000	-121000.0	495000.0	75	0.775114
age	11.0000	25.0000	14.000	-10.0	46.0	75	0.775114

Figure 6: Outlier detection for numerical variables.

Outliers were identified using the IQR method. Most numeric variables had fewer than 1% outliers, except for price (3.9%), while mileage and age had 0.78%, reflecting very old or high-mileage vehicles. Handling outliers is unnecessary, as they are few and meaningful, representing real world variations. Therefore, removing them could discard important information about luxury, rare, or unusually used cars.

Variable Correlation & Multi-Collinearity



	Feature	GVIF	DF	GVIF^(1/(2*DF))
0	air_condition	0.930124	1	0.964430
1	brand_grouped	0.050323	13	0.891390
2	power_steering	0.636402	1	0.797748
3	fuel_type	0.215341	3	0.774203
4	gear	0.527885	1	0.726557
5	power_mirror	0.495554	1	0.703956
6	power_window	0.443444	1	0.665916
7	engine_cc	0.417772	1	0.646353
8	age	0.391692	1	0.625853

Figure 7: Correlation matrix and GVIF values for explanatory variables.

The correlation matrix shows strong positive correlation between vehicle age and mileage while price is moderately positively correlated with YOM and negatively with age and mileage. This indicates newer and less-used vehicles command higher prices. All variables have low adjusted GVIF values, including high-degree-of-freedom factors like brand_grouped and fuel_type, indicating no multicollinearity. Overall, the variables are sufficiently independent and suitable for further analysis without distortion.

Objective 1: Brand Equity and Depreciation Analysis

This objective investigates how brand, age, and engine capacity influence second-hand vehicle prices, quantifying brand premiums, mapping depreciation, and isolating the effects of performance and age. While visualizations help explore the data, they cannot fully reveal conditional or non-linear patterns. An initial OLS model assessed linear relationships, but residual analysis showed it was insufficient. A Generalized Additive Model was therefore used to capture non-linear effects and categorical differences while controlling for other covariates.

Since **age** and **mileage** were perfectly correlated, we applied a **residualization approach** to capture unique mileage data not explained by age; however, this showed no improvement, so we retained only **age** to ensure a cleaner model fit.

Vehicle models were excluded to avoid high-cardinality issues, with engine capacity (engine_cc) used instead as a parsimonious proxy for performance and market positioning. This continuous variable captures model based price variations while allowing analysis of both linear and non-linear effects. Brand was retained to account for baseline price differences driven by reputation and perceived reliability factors that technical specifications alone cannot explain.

For this analysis, only vehicle age, engine_cc, and brand were selected because they represent the primary factors considered by buyers. Age reflects depreciation, engine_cc proxies for model and performance differences, and brand captures manufacturer-level price variation. Other variables were excluded to maintain interpretability and avoid high-dimensional noise, as the goal is exploratory analysis rather than full prediction.

Ordinary Least Squares

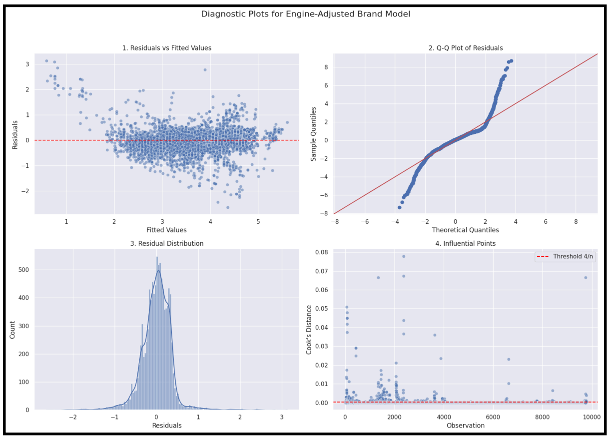


Figure 8: Residual diagnostics for the OLS regression model.

An initial OLS model was fitted and its residuals were examined. While the residual histogram suggested approximate normality, the QQ plot showed tail deviations, which was confirmed by a significant Anderson Darling test. The residuals versus fitted plot appeared homoscedastic, yet the Breusch Pagan test indicated the presence of heteroscedasticity. Given the contradicting results we can't consider the OLS coefficient as valid measurements for hypothesis testing.
Therefore, non parametric methods were used for hypothesis testing as they do not rely on restrict assumptions.

Kruskal - Wallis Test for Brand price differences

To test whether car prices differ across the brands, we used Kruskal Wallis test on the log transformed price (to reduce skewness).

- **Null hypothesis (H_0):** The distribution of price_transformed is the same across all car brands.
- **Alternative hypothesis (H_1):** At least one brand has a different price distribution.

Results: Kruskal-Wallis H statistic = 2682.46 , **P-value = 0** < 0.05

Since the p-value is far below 0.05, we reject H_0, indicating that price distributions differ significantly between at least some brands.

Pairwise Brand Comparisons Using Mann-Whitney U Test

To investigate which car brands differ from Toyota in terms of log-transformed prices, **pairwise Mann-Whitney U tests** were conducted. Since multiple comparisons were performed, p-values were adjusted using the **Bonferroni correction** to control for false positives.

- **Null hypothesis (H_0):** The price distribution of Toyota is equal to that of the other brand.

- **Alternative hypothesis (H_1):** The price distribution of Toyota differs from that of the other brand.

After correction, almost all brands showed significant differences from Toyota. This indicates that, controlling for multiple tests, most brands have price distributions that are statistically distinct from Toyota.

Directional Pairwise Comparisons Against Toyota

To assess not only whether other brands differ from Toyota, but also the **direction of the difference**, one-sided Mann-Whitney U tests were performed for the most common brands in Sri Lanka.

- **Null hypothesis (H_0):** Prices of Toyota are not greater (or not smaller) than the other brand.

- **Alternative hypothesis (H_1):** Prices of Toyota are greater (or smaller) than the other brand.

The one sided Mann Whitney U tests revealed significant price differences between Toyota and other common brands in Sri Lanka with clear directional trends. Toyota vehicles are significantly less expensive compared to luxury brands such as **Mercedes Benz and Honda**. But when compared with other budget or entry level brands such as **Nissan, Suzuki, Mitsubishi and Tata**, Toyota is more expensive. These results confirm the expected brand positioning in the market and also quantify the price differences.

Partial Spearman correlation for Price and Engine Capacity

To investigate the effect of engine capacity on vehicle price while accounting for the influence of age and brand, a Spearman's partial correlation analysis was conducted. This approach allows us to examine the relationship between engine size and price independently of other confounding variables.

- **Null (H_0):** There is no monotonic association between engine size and price after controlling for age and brand.

- **Alternative (H_1):** There is a monotonic association between engine size and price after controlling for age and brand.

The results indicated a moderate positive partial correlation (ρ=0.370, 95% CI: 0.35–0.39, p = 0 <0.05), suggesting that larger engine capacity contributes to higher vehicle prices independently of age and brand. This finding confirms that engine size is an important determinant of second-hand vehicle pricing, beyond the effects of depreciation and brand differences.

Partial Spearman correlation for Price and age

To examine the effect of vehicle age on price while controlling for engine capacity and brand, a Spearman's partial correlation analysis was performed. This method allows us to isolate the relationship between age and price, removing the confounding effects of other key variables.

- **Null (H_0):** There is no monotonic association between age and price after controlling for engine capacity and brand.

- **Alternative (H_1):** There is a monotonic association between age and price after controlling for engine capacity and brand.

The analysis revealed a strong negative partial correlation ($\rho = -0.814$, 95% CI: [-0.82, -0.81], $p = 0 < 0.05$), indicating that older vehicles are consistently priced lower, independent of engine size and brand. This result quantifies the impact of depreciation, confirming that age is a major determinant of second-hand vehicle pricing.

To further explore the relationships identified through correlation and non-parametric tests, a **Generalized Additive Model (GAM)** was fitted. Unlike linear models or simple correlations, GAMs allow for non-linear, smooth effects of continuous predictors (e.g. age and engine capacity) while also accommodating categorical factors such as brand.

Generalized Additive Model (GAM)

```
LinearGAM
=============================================== ==========================================================
Distribution:                        NormalDist Effective DoF:                                     44.5175
Link Function:                     IdentityLink Log Likelihood:                                 -3945.2049
Number of Samples:                         9770 AIC:                                             7981.4446
                                                AICc:                                            7981.8802
                                                GCV:                                                 0.133
                                                Scale:                                              0.3632
                                                Pseudo R-Squared:                                   0.7647
==========================================================================================================
Feature Function                  Lambda               Rank         EDoF         P > x        Sig. Code
================================= ==================== ============ ============ ============ ============
s(0)                              [0.6]                20           14.0         1.11e-16     ***
f(2)                              [0.6]                14           12.9         1.11e-16     ***
s(1)                              [0.6]                20           17.6         1.11e-16     ***
intercept                                              1            0.0          1.11e-16     ***
==========================================================================================================
```

Figure 9: Generalized Additive Model (GAM) results.

The GAM model was fitted by taking price (log transformed) as the target variable and brand, age and engine_cc(log transformed) as the predictors.

(Since the python library have limitations on GAM, R was used for model adequacy checking and the relevant codes are in the notebook file as a text)

For the brand we take Toyota as the reference brand as it is a common car brand in Sri Lanka and is suitable for comparison with other car brands.

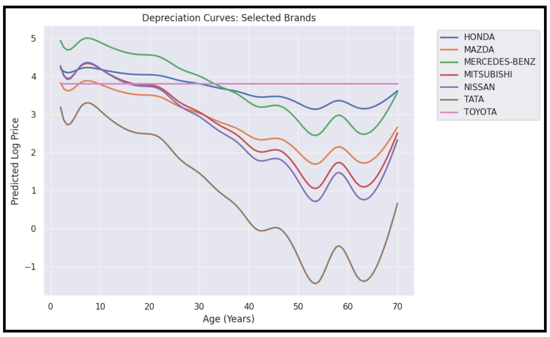


Figure 10: Brand-wise depreciation with age relative to Toyota.

How do brands depreciate with age?

In Sri Lanka's automotive market, **Toyota** serves as the primary benchmark for value retention, largely due to its reputation for reliability and the increased demand driven by import restrictions. While most brands depreciate relative to this baseline, **Tata** exhibits the most severe value loss as a budget-tier option. Among the common Japanese alternatives, **Nissan**, **Mitsubishi**, and **Mazda** all show steeper depreciation curves than Toyota; however, **Mazda** retains value more effectively than its mid-tier peers, likely owing to its "semi-luxury" brand perception.

In the premium segment, **Mercedes-Benz** faces significant relative depreciation as vehicles age, reflecting the high maintenance costs and diminished demand for aging luxury assets. Interestingly, **Honda** maintains a trajectory more closely aligned with Toyota than the European luxury brands, signaling a strong market preference for Japanese engineering. Ultimately, Toyota's dominance as the reference point highlights a market where sellers leverage high demand and vehicle scarcity to maintain premium pricing across all age brackets.

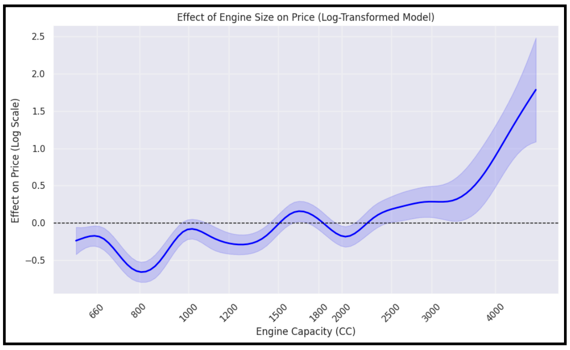


Figure 11: Price variation by engine capacity.

While the plot confirms a positive correlation between engine_cc and price, engine capacity itself is not the sole driver of value. It acts as a proxy for the vehicle's broader segment, Smaller engine vehicles typically belong to budget friendly vehicles where price sensitivity is high. As capacity increases beyond 1500cc or 2000cc, we are moving into the territory of sedans and luxury SUVs. In these categories, higher price is not just paying for the engine, but for premium build quality, advanced technology and comfort features. Thus, the upward curve captures the compound value of both engine's power and the luxury status it presents.

Objective 2 : Mechanical performance Analysis

The purpose of this objective is to understand and identify how key mechanical characteristics of cars are associated with their market price in Sri Lanka's second-hand car market.Analyzing mechanical specifications is important because it provides insights into which mechanical features buyers value most and helps sellers to price fairly.
In this project fuel type,transmission (gear) type and engine capacity are considered as the primary mechanical features.These features were chosen as the mechanical features because they directly influence vehicle performance and buyer preference.These features capture the key mechanical aspects that determine second- hand car prices in Sri Lanka. The analysis will quantify how each of these features is associated with price.

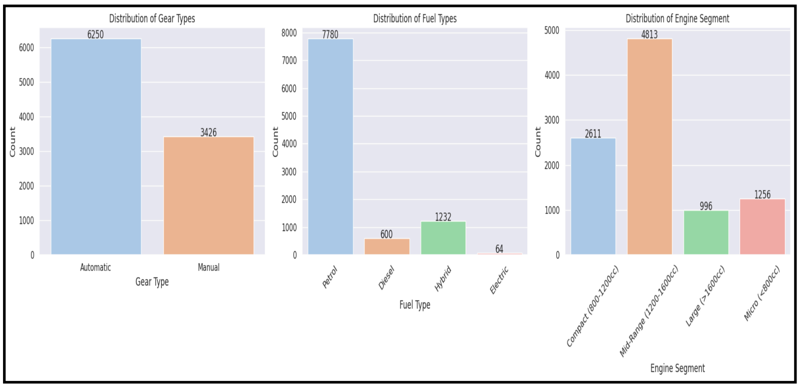


Figure 12: Number of vehicles by fuel type, engine segment, and gear type.

This dataset contains the 4 fuel types.Petrol cars are the most common in the market, while there are only a few electric cars.Among the Automatic and Manual cars,Automatic cars dominate the market.Mid range engine size cars are most common in the market compared to the other engine segments.

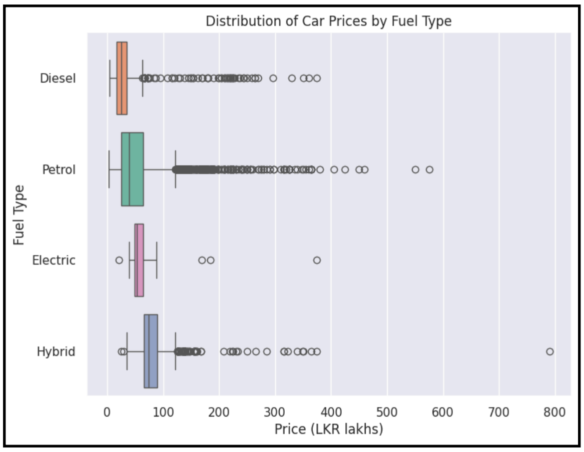


Figure 13: Distribution of car prices across different fuel types.

The analysis of car price by fuel type (***Figure 13***) reveals that Hybrid cars have the highest median price and the widest price range,including an extreme value near 790 LKR lakhs which is unusually expensive compared to most Hybrid cars.Electric cars have the second highest median price with tighter price range.Petrol and Diesel have similar lower median prices compared to Hybrid and Electric vehicles.However,among the Petrol cars ,there are more very expensive cars than Diesel cars.

For this analysis,vehicles were grouped into Traditional (Petrol and Diesel) and Alternative (Hybrid and Electric) car groups.This grouping was used to clearly show market differences.

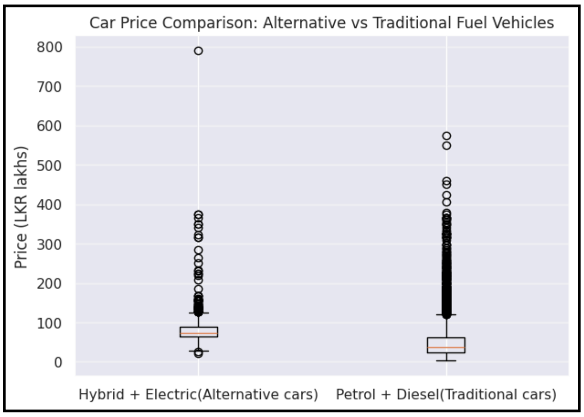


Figure 14: Distribution of car prices for alternative and traditional fuel vehicles.

This analysis (***Figure 14***) shows that Alternative car types have higher median prices and a wider upper price range compared to the traditional car types.Both types shows right skewed distributions with high prices.These distributions clearly show that,within the Hybrid and Alternative fuel categories,there is a car(Toyota LEXUS 600HL) with an high price of 790 lakhs LKR compared to other cars in the same categories.This extreme prices reflects distorted or inflated pricing during the closed market period in Sri Lanka,rather than typical market conditions.

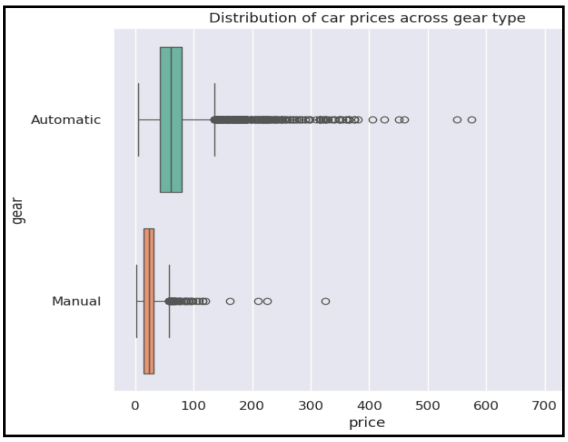


Figure 15: Distribution of car prices by gear type.

This boxplot (***Figure 15***) shows clear price differences between Automatic and Manual gear type cars in the Sri Lankan second hand market.Automatic cars show a higher median price than manual cars.Both gear types show right skewed distributions with high prices,However Automatic vehicles display more extreme prices reaching the highest price levels in the dataset. In contrast,Manual car types are at lower price levels with fewer extreme values.

The boxplot (***Figure 16***) analysis was conducted to examine the distribution of car prices (in LKR Lakhs) across four engine capacity segments.This shows car prices are right-skewed across all engine segments, with a clear long-tail pattern driven by high-value vehicles.The **Micro (<800cc)** segment shows a slightly higher median price compared to other engine segments.Although the **Large**

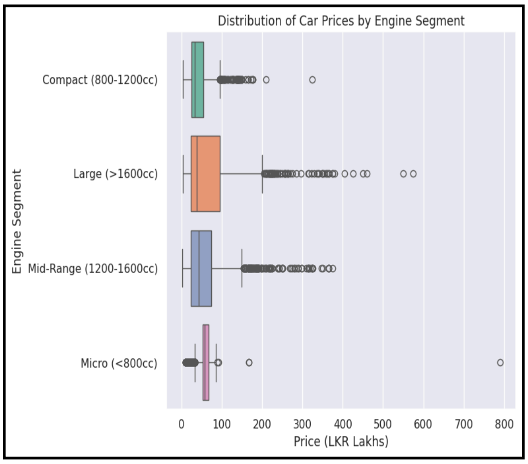


Figure 16: Distribution of car prices by engine capacity segment.

(>1600cc) segment exhibits the greatest variability and most extreme high values but its median price is lower than compared to micro segment.

To understand the relationships between car prices and key mechanical characteristics,advanced statistical methods were applied to answer the following research questions.Since the violations of normality and homoscedasticity of car prices across fuel type,gear type and engine segments ,non parametric tests such as Kruskal-Wallis, Mann-Whitney tests and post hoc(Dunn's test with Bonferroni correction and Holm correction) tests were used to test the hypotheses.

- **Do car prices differ across fuel types (Petrol, Diesel, Hybrid, Electric)?**

To test whether used car prices differ across fuel types,the Kruskal Wallis test was applied to prices due to unequal variances across the groups .
H_0: Median vehicle prices are equal across fuel types (Petrol, Diesel, Hybrid, Electric).
H_1: At least one fuel type has a different median price.
The test provided sufficient evidence of a significant difference in median prices across fuel types (H = 1606.22,p =0.00) at 5% level of significance. Post hoc test (Dunn's test) was used to identify which fuel type pairs differ .
H_0:Median prices of the two fuel types are equal
H_1:Median prices of the two fuel types are different

	Comparison	Adjusted p-value	Significant?
0	Petrol vs Diesel	0.0000	Yes
1	Petrol vs Hybrid	0.0000	Yes
2	Petrol vs Electric	0.0001	Yes
3	Diesel vs Hybrid	0.0000	Yes
4	Diesel vs Electric	0.0000	Yes
5	Hybrid vs Electric	0.0001	Yes

Figure 17: Dunn's post hoc test results for pairwise comparisons of vehicle prices across fuel types.

All pairwise comparisons were statistically significant at 5% significance level.That means the median prices of all fuel types differ from each other at 5% significance level.(Both Bonferroni correction and Holm correction methods consistently showed that all pairwise comparisons were statistically significant at the 5% level.)

- **Do Alternative fuel cars (Hybrid and Electric) command higher prices than Traditional fuel cars (Petrol and Diesel) in the second hand car market?**

```
Mann Whitney U = 8944475.5
One-sided p-value = 6.650198346477350-309
```

The non parametric Mann Whitney U (one sided) test was used to compare median prices between the Alternative and Traditional car groups due to non normal price distribution and unequal variances across the groups.
H_0:Median prices of alternative fuel cars are equal to those of traditional fuel cars.
H_1:Median prices of alternative fuel cars are higher than those of traditional fuel cars.
The test provided a p value effectively equal to zero (6.65×10^{-309}) far below the 5% significance level.This provides strong evidence to reject H_0 indicating that alternative fuel cars have higher prices reflecting a clear market premium for hybrid and electric cars.

Above results suggest that used car prices are strongly influenced by fuel type.Electric and Hybrid cars command higher prices while Petrol and Diesel are comparatively lower priced.This highlights clear differences in pricing strategies across fuel types.

- **How transmission(gear) type affects price and whether Automatic cars are priced higher than Manual cars due to driving convenience?**

The Mann Whitney U (one sided) test was applied to examine whether gear type affects the used car prices ,considering driving convenience as a factor.

H_0: Median prices of Automatic and Manual vehicles are equal.
H_1: Automatic vehicles have higher median prices than Manual vehicles.

The test yielded a **Mann Whitney U statistic of 19,787,379.5** with a **p-value =0.00**. At the 5% significance level, this provides strong evidence to reject H_0 indicating that automatic vehicles are priced higher than manual cars .This result suggests that transmission(gear) type is associated with the car price because it reflects driving convenience.Automatic cars are easier to drive in traffic areas and are simpler for new drivers to use compared to manual cars. This added convenience contributes to their higher market prices in the second-hand car market.

- **Do Vehicle Prices Differ Across Engine Types?**

	Comparison	p-value	Significant?
0	Compact (800-1200cc) vs Mid-Range (1200-1600cc)	1.932567e-29	Yes
1	Compact (800-1200cc) vs Large (>1600cc)	1.716977e-19	Yes
2	Compact (800-1200cc) vs Micro (<800cc)	1.701413e-70	Yes
3	Mid-Range (1200-1600cc) vs Large (>1600cc)	3.502945e-01	No
4	Mid-Range (1200-1600cc) vs Micro (<800cc)	2.122961e-25	Yes
5	Large (>1600cc) vs Micro (<800cc)	1.283856e-09	Yes

Figure 18: Dunn's post hoc test for pairwise comparisons of vehicle prices across engine types.

To examine whether vehicle prices differ across engine types, a Kruskal Wallis test was conducted due to non-normal price distributions and unequal variances across engine segments.

H_0: The median vehicle price is the same across all engine types.
H_1: At least one engine type has a different median vehicle price.

The test provided sufficient evidence of a significant difference in median prices across engine segments. ($H = 340.979$, $p = 0.00$) at 5% level of significance.Post hoc test (Dunn's test with Bonferroni correction) was used to identify which engine segment pairs differ.Mid-range (1200–1600cc) and Large (>1600cc) vehicles do not show a statistically significant difference in median prices, while all other engine-type pairs differ significantly at 5% level of significance.

Objective 3: Comfort Feature Analysis

This objective evaluates how Air Conditioning, Power Steering, Power Mirrors, and Power Windows influence prices in Sri Lanka's secondary market in the given period. By visualizing long-term trends and using statistical testing, it determines if these features still command a higher price or have become basic expectations. The analysis identifies shifts across vehicle eras and quantifies the extra cost associated with each feature, providing evidence of how feature rich vehicles went against standard depreciation during this period.

Count plots in ***Figure 19*** illustrate the frequency distribution of primary comfort features across the dataset. The data indicates that Air Conditioning (AC) has reached near-total market saturation, present in over 95% of listings. Power Steering, Power Windows, and Power Mirrors also show high frequency, suggesting these are now baseline expectations for consumers rather than premium add-ons in the Sri Lankan secondary market.

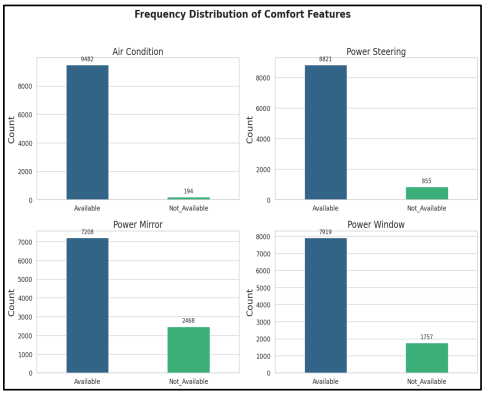


Figure 19: Frequency Distribution of Comfort Features

The clustered column chart in ***Figure 20*** depicts the percentage penetration of features across the three defined eras. While AC was a staple even in the "Old" category, there is a marked "technological leap" in the "Intermediate" category for Power Mirrors and Windows. By the "Modern" category, the gap between these features has closed, signifying that the market has transitioned into a high feature baseline where basic models are increasingly rare.

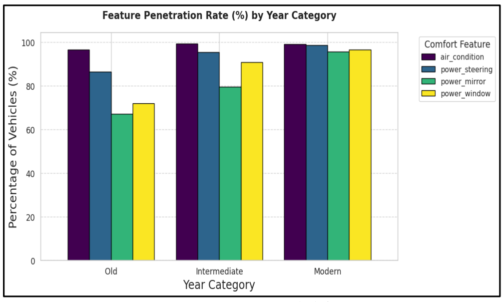


Figure 20: Feature Penetration Rate (%) by Year Category

Box plots in ***Figure 21*** reveal a higher correlation between the presence of comfort features and vehicle valuation. Vehicles equipped with a full suite of power features consistently maintain higher median prices. Notably, the "Available" categories exhibit a wider range of high-value outliers, likely representing luxury segments where these features are standardized alongside higher engine capacities.

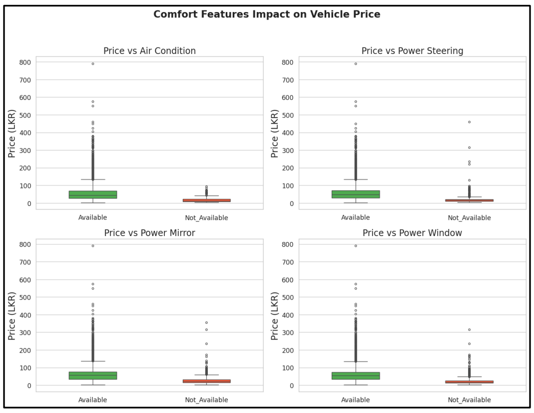


Figure 21: Comfort Features Impact on Vehicle Price

The strip plots in ***Figure 22*** provide a granular view of how pricing "abnormality" manifests across different manufacturing periods. While the "Old" category (pre-2010) shows stagnant pricing regardless of features, the "Intermediate" (2010–2017) and "Modern" (2018+) categories display massive price variance. This confirms that comfort features serve as a significant multiplier for the appreciation trends observed during the import restricted period.

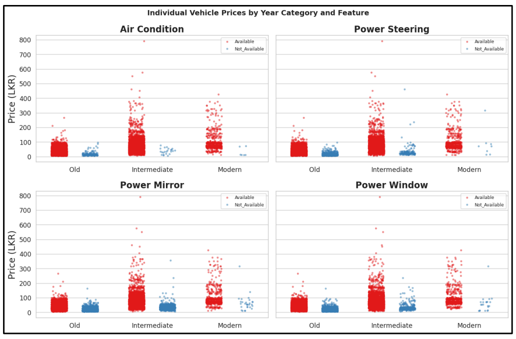


Figure 22: Individual Vehicle Prices by Year Category and Feature

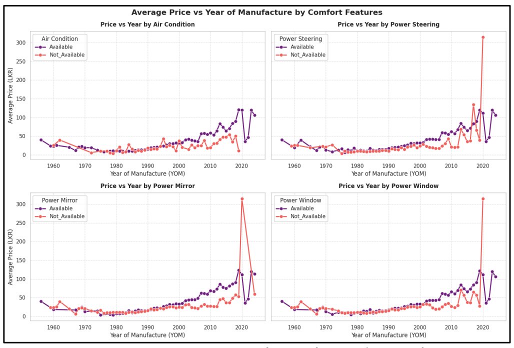


Figure 23: Average Price vs Year of Manufacture by Comfort Features

The line charts in ***Figure 23*** track average price against the Year of Manufacture (YOM). Across all categories, there is a steady increase in average price for vehicles manufactured after the year 2000. There is a significant, sharp spike in the price of vehicles without Power Steering, Power Mirrors, and Power Windows around the year 2020. The data shows significant price fluctuations in the most recent years (2020–2024), indicating a volatile market for newer vehicles.

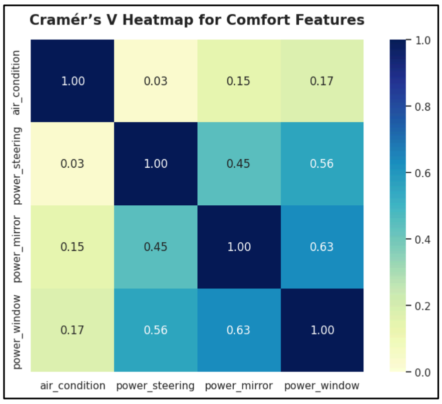


Figure 24: Cramér's V Heatmap for Comfort Features

The Cramér's V heatmap in ***Figure 24*** quantifies the relationships between categorical variables. A strong association (V≅0.63) exists between Power Windows and Power Mirrors, suggesting these features are typically bundled. In contrast, Air Conditioning shows a negligible correlation with other features (ranging from 0.03 to 0.17), highlighting its status as a universal standard across all vehicle segments.

Does price significantly vary based on the availability of comfort features?

Test	Variable	Statistic	p-value	Conclusion
Normality (D'Agostino K²)	Price	7,493.8353	0.0000e+00	Not Normal
Levene's Test	Air Condition	51.7698	6.7041e-13	Unequal Variances
Levene's Test	Power Steering	237.8043	5.0491e-53	Unequal Variances
Levene's Test	Power Mirror	522.5523	1.0996e-112	Unequal Variances
Levene's Test	Power Window	418.1072	5.1980e-91	Unequal Variances

Figure 25: Normality for Price(D'Agostino K^2) and Levene's Test per Comfort Feature

Initial normality testing using the **D'Agostino K^2 test** (as the Shapiro-Wilk test is unreliable for large datasets) yielded a p-value of 0.000 as in ***Figure 25***, indicating that the price distribution significantly deviates from normality and the **Levene's test** for each comfort feature results unequal variances. Because assumptions are violated, the analysis proceeded using non-parametric methods, specifically the **Mann-Whitney U test**, to evaluate price differences for availability of each comfort feature. The hypothesis for each comfort feature (i = Air Conditioning, Power Steering, Power Mirror, Power Window) was defined as follows:

Test	Variable	Statistic	p-value	Conclusion
Mann–Whitney U	Air Condition	1,552,860	1.0241e-60	Significant
Mann–Whitney U	Power Steering	6,651,322	1.3266e-298	Significant
Mann–Whitney U	Power Mirror	15,247,108	0.0000e+00	Significant
Mann–Whitney U	Power Window	12,250,961	0.0000e+00	Significant

Figure 26: Mann–Whitney U test (two-sided) results

H_0: There is no significant difference in the median price between vehicles with feature i available and those without feature i available.
H_1: There is a statistically significant difference in the median price between vehicles with feature i available and those without feature i available.

The U Statistics for all four features returned p-values effectively at near zero ($p < 0.05$) as in ***Figure 26***, leading to the rejection of the null hypothesis in every case. This confirms that the price premiums associated with available comfort features are statistically significant and are not a result of random variation.

Objective 4 : Market Sentiment and Geo economic Analysis

This objective aims to examine market sentiment and geo-economic patterns within the vehicle marketplace by assessing whether geographic location and financial accessibility influence vehicle pricing and availability. Specifically, the analysis investigates differences between Urban and Non-Urban markets and evaluates the role of leasing as a financial mechanism in shaping market access. By controlling for inherent variation in vehicle types through stratified analysis, the objective seeks to determine whether observed price and availability differences reflect genuine geographic and financial segmentation or an integrated national market structure.

Classification of Urban and Non-Urban Locations

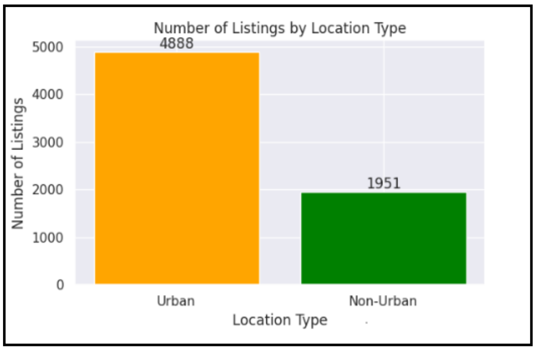


Figure 27: Distribution of vehicle listings by location type.

The distribution of listings based on location type, reveals an overwhelming dominance of vehicles within the Urban type of location, with the share of Non-Urban locations being substantially smaller. This disparity reveals that the supply of vehicles is highly centralized, with Urban locations acting as the main hubs for providing vehicles. The larger share of vehicles within Urban locations reveals that these areas are more active and accessible.

Further, the distribution of the number of listings with respect to leasing availability shows the strong dominance of non-leasing vehicles within the market. Most of the listings are for non-leasing options, while only a very small fraction of the vehicles are available for lease.

Figure 28: Distribution of vehicle listings by leasing availability.

Price Trends by Year of Manufacture Across Location Types

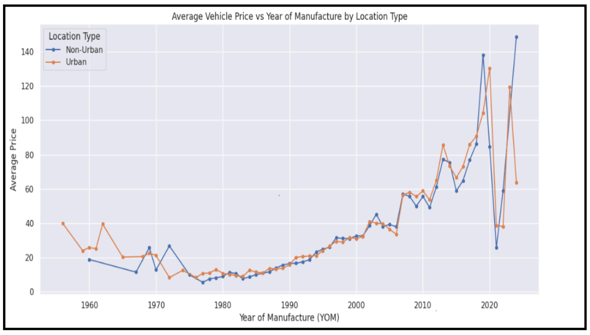


Figure 29: Average vehicle price by year of manufacture for Urban and Non-Urban markets.

The relationship between the Average Vehicle Price and the Year of Manufacture has a strong upward trend for the Urban and Non-Urban markets, which implies that newer vehicles are always associated with higher prices regardless of the market type. The two curves are almost identical over the entire range of the year of manufacture, with minor oscillations between the two markets. Overall, the conclusion that the market is substantially integrated and that vehicle age, rather than geographic considerations, drives pricing is supported by the uniformity in price trends across locales.

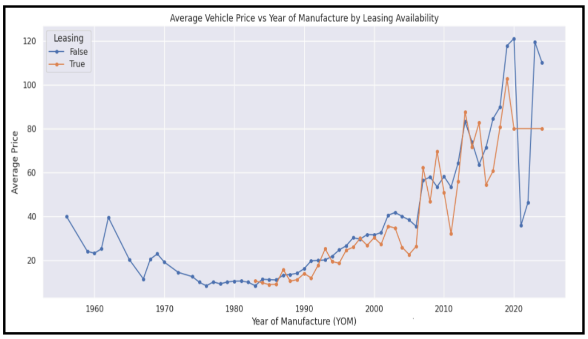


Figure 30: Average vehicle prices by year of manufacture, grouped by leasing availability.

This shows the average price of vehicles based on different years of manufacture, distinguishing between those available for leasing and those not available for leasing. For both leased and non-leased vehicles, it is evident that the price rises steadily in line with the year of manufacture, suggesting that the year of manufacture is one of the primary factors in determining the price.

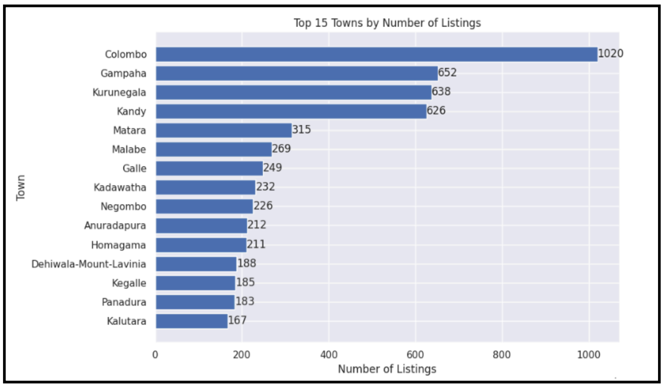


Figure 31: Top fifteen towns by number of vehicle listings.

The chart highlights the top fifteen towns by number of vehicle listings, showing a strong concentration of market activity in a small number of urban centers. Colombo dominates the market by a substantial margin, followed by Gampaha, Kurunegala, and Kandy. These towns account for a disproportionately large share of total listings compared to other locations.

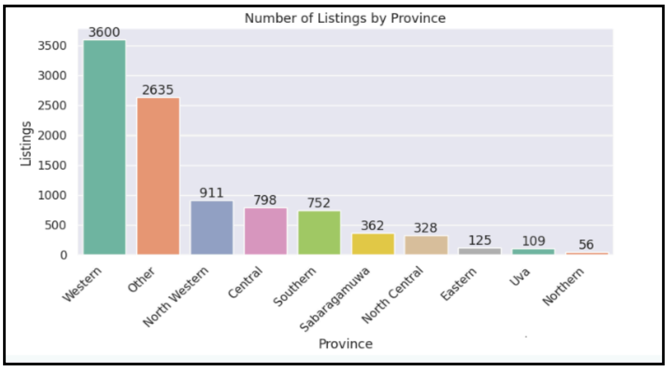


Figure 32: Distribution of vehicle listings by province.

shows a high level of concentration in terms of geographical location. The Western Province dominates in terms of listings, followed by "Other," which comprises towns not identified with any particular province. Next are North Western, Central, and Southern provinces in terms of moderate levels of listings. At the least represented end of the spectrum are Eastern, Uva, and Northern provinces.

Do vehicle prices differ across listed locations ?

To evaluate whether vehicle prices differ across listed locations, a non-parametric bootstrap approach was used to estimate the difference in median prices between Urban and Non-Urban listings. Prices within each group were resampled with replacement over 10,000 iterations to generate a bootstrap distribution of the median difference, from which a 95% confidence interval was constructed.

H_0: $\text{Median}_{(\text{Urban})} - \text{Median}_{(\text{Non-Urban})} = 0$ **H_1: $\text{Median}_{(\text{Urban})} - \text{Median}_{(\text{Non-Urban})} \neq 0$**

```
95% Confidence Interval for Difference in Medians (Urban - Non-Urban): (-0.5746, 3.9500)
Mean Bootstrapped Difference: 1.5691
```

The 95% bootstrap confidence interval: (−0.57, 3.95) includes zero, indicating no statistically significant price difference between the two location groups. Although the average difference is positive, it is not statistically meaningful.

Do vehicle prices differ across leasing availability ?

To examine whether leasing availability affects vehicle prices, the same method we used under the listed location is used.

H_0: $\text{Median}_{(\text{Leasing})} - \text{Median}_{(\text{Leasing}))} = 0$ **H_1: $\text{Median}_{(\text{Leasing})} - \text{Median}_{(\text{Leasing})} \neq 0$**

```
95% Confidence Interval for Difference in Medians (Leasing - No Leasing): (-11.3375, -3.7500)
Mean Bootstrapped Difference: -7.5622
```

The 95% bootstrap confidence interval: (−11.34, −3.75) does not include zero, indicating a statistically significant difference. The negative interval suggests that vehicles with leasing availability are typically priced lower than non-leasing vehicles.

Bootstrap Analysis of Price Differences by Location and Leasing Availability

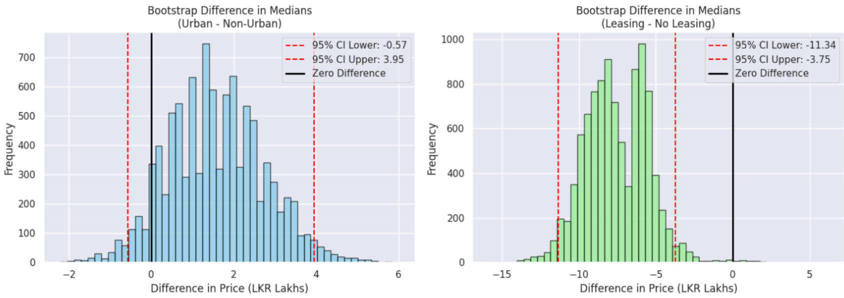
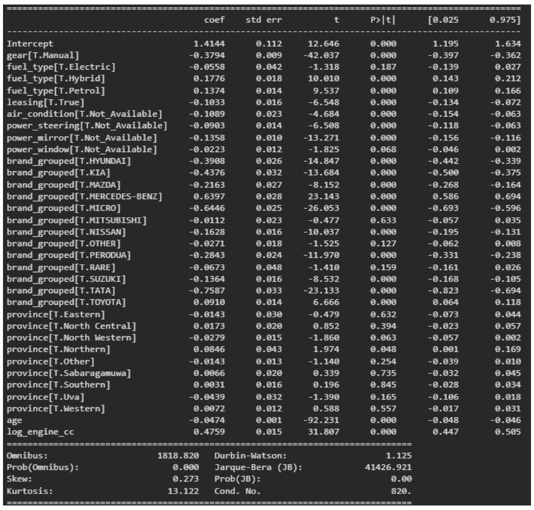


Figure 33: Bootstrap Analysis of Price Differences by Location and Leasing Availability

The bootstrap distributions illustrate the estimated differences in median vehicle prices across location type and leasing availability. For Urban versus Non-Urban listings, the 95% confidence interval spans zero, indicating no statistically significant difference in median prices. In contrast, for Leasing versus No Leasing listings, the entire confidence interval lies below zero, demonstrating a statistically significant lower median price for vehicles with leasing availability.

Modeling



	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	1.4144	0.112	12.646	0.000	1.195	1.634
gear[T.Manual]	-0.3794	0.009	-42.037	0.000	-0.397	-0.362
fuel_type[T.Electric]	-0.0558	0.042	-1.318	0.187	-0.139	0.027
fuel_type[T.Hybrid]	0.1776	0.018	10.010	0.000	0.143	0.212
fuel_type[T.Petrol]	0.1374	0.014	9.537	0.000	0.109	0.166
leasing[T.True]	-0.1033	0.016	-6.548	0.000	-0.134	-0.072
air_condition[T.Not_Available]	-0.1089	0.023	-4.684	0.000	-0.154	-0.063
power_steering[T.Not_Available]	-0.0903	0.014	-6.508	0.000	-0.118	-0.063
power_mirror[T.Not_Available]	-0.1358	0.010	-13.271	0.000	-0.156	-0.116
power_window[T.Not_Available]	-0.0223	0.012	-1.825	0.068	-0.046	0.002
brand_grouped[T.HYUNDAI]	-0.3908	0.026	-14.847	0.000	-0.442	-0.339
brand_grouped[T.KIA]	-0.4376	0.032	-13.684	0.000	-0.500	-0.375
brand_grouped[T.MAZDA]	-0.2163	0.027	-8.152	0.000	-0.268	-0.164
brand_grouped[T.MERCEDES-BENZ]	0.6397	0.028	23.143	0.000	0.586	0.694
brand_grouped[T.MICRO]	-0.6446	0.025	-26.053	0.000	-0.693	-0.596
brand_grouped[T.MITSUBISHI]	-0.0112	0.023	-0.477	0.633	-0.057	0.035
brand_grouped[T.NISSAN]	-0.1628	0.016	-10.037	0.000	-0.195	-0.131
brand_grouped[T.OTHER]	-0.0271	0.018	-1.525	0.127	-0.062	0.008
brand_grouped[T.PERODUA]	-0.2843	0.024	-11.970	0.000	-0.331	-0.238
brand_grouped[T.RARE]	-0.0673	0.048	-1.410	0.159	-0.161	0.026
brand_grouped[T.SUZUKI]	-0.1364	0.016	-8.532	0.000	-0.168	-0.105
brand_grouped[T.TATA]	-0.7587	0.033	-23.133	0.000	-0.823	-0.694
brand_grouped[T.TOYOTA]	0.0910	0.014	6.666	0.000	0.064	0.118
province[T.Eastern]	-0.0143	0.030	-0.479	0.632	-0.073	0.044
province[T.North Central]	0.0173	0.020	0.852	0.394	-0.023	0.057
province[T.North Western]	-0.0279	0.015	-1.860	0.063	-0.057	0.002
province[T.Northern]	0.0846	0.043	1.974	0.048	0.001	0.169
province[T.Other]	-0.0143	0.013	-1.140	0.254	-0.039	0.010
province[T.Sabaragamuwa]	0.0066	0.020	0.339	0.735	-0.032	0.045
province[T.Southern]	0.0031	0.016	0.196	0.845	-0.028	0.034
province[T.Uva]	-0.0439	0.032	-1.390	0.165	-0.106	0.018
province[T.Western]	0.0072	0.012	0.588	0.557	-0.017	0.031
age	-0.0474	0.001	-92.231	0.000	-0.048	-0.046
log_engine_cc	0.4759	0.015	31.807	0.000	0.447	0.505

Omnibus:	1818.820	Durbin-Watson:	1.125
Prob(Omnibus):	0.000	Jarque-Bera (JB):	41426.921
Skew:	0.273	Prob(JB):	0.00
Kurtosis:	13.122	Cond. No.	820.

Figure 34: Linear Regression Model (OLS)

The analysis emphasizes statistical inference over prediction to quantify how vehicle characteristics influence second-hand prices. OLS regression was first employed as a baseline due to its interpretability and strong theoretical foundation. However, the initial model using raw variables violated normality and homoscedasticity assumptions. Log transformations of price and engine capacity were therefore applied, improving residual behavior and stabilizing variance.

Log transformations stabilize variance and normalize price skewness, improving statistical inference. Log-transforming **engine capacity** captures diminishing marginal effects—where size impacts price proportionally rather than additively. These adjustments ensure healthier residuals and allow for more intuitive, elasticity-based interpretations. Model diagnostics show mixed results: the **Q-Q plot** reveals departures from the

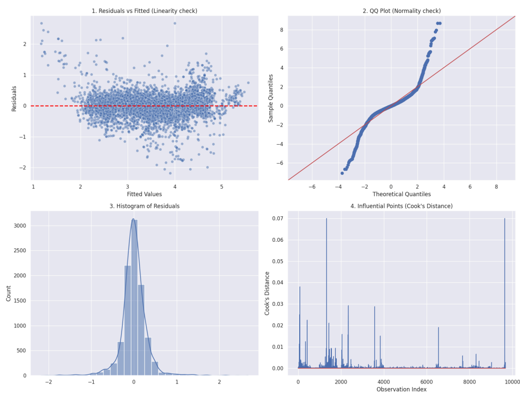


Figure 35: Residual Analysis of OLS

diagonal, indicating residuals are not perfectly normal. However, the **Residuals vs. Fitted** plot lacks "fan" or "funnel" patterns, suggesting the variance is constant and the assumption of **homoscedasticity** holds. However, statistical tests challenge visual diagnostics. The **Shapiro-Wilk** test ($p = 0 < 0.05$) confirms the non-normality suggested by the Q-Q plot. More importantly, the **Breusch-Pagan** test ($p = 0 < 0.05$) rejects the null hypothesis of homoscedasticity, revealing statistically significant variance in errors that the visual inspection failed to capture.

Transition to GAM

Given the violations of normality and heteroscedasticity, OLS assumptions do not hold for this dataset. This indicates a non-linear relationship between price and its predictors. OLS attempts to fit straight lines to inherently non-linear market trends, such as depreciation, which explains the observed errors.

Why GAM?

The Generalized Additive Model (GAM) was chosen for its ability to capture non-linear patterns—such as rapid early depreciation and late-life price plateaus—that standard linear regression misses. Unlike “black-box” methods, GAM retains an additive structure, balancing flexibility and interpretability by allowing clear visualization of each feature’s effect on price while improving accuracy over OLS.

Generalized Linear Model (GAM)

```
LinearGAM
=============================================== ==========================================================
Distribution:                      NormalDist Effective DoF:                                       58.9253
Link Function:                   IdentityLink Log Likelihood:                                   -1360.3261
Number of Samples:                       9676 AIC:                                               2840.5027
                                              AICc:                                              2841.2621
                                              GCV:                                                  0.0789
                                              Scale:                                                0.2793
                                              Pseudo R-Squared:                                     0.8501
==========================================================================================================
Feature Function                  Lambda               Rank         EDoF         P > x        Sig. Code
================================= ==================== ============ ============ ============ ============
s(0)                              [0.6]                20           14.2         1.11e-16     ***
s(1)                              [0.6]                20           13.9         1.11e-16     ***
f(2)                              [0.6]                2            1.0          1.11e-16     ***
f(3)                              [0.6]                4            3.0          1.11e-16     ***
f(4)                              [0.6]                10           9.0          1.65e-01
f(5)                              [0.6]                14           12.9         1.11e-16     ***
l(6)                              [0.6]                1            1.0          9.84e-12     ***
l(7)                              [0.6]                1            1.0          4.74e-08     ***
l(8)                              [0.6]                1            1.0          5.51e-07     ***
l(9)                              [0.6]                1            1.0          1.11e-16     ***
l(10)                             [0.6]                1            1.0          5.67e-01
intercept                                              1            0.0          1.11e-16     ***
==========================================================================================================
```

```
Index     Function     Feature Name
-------------------------------------
0         s(0)         log_engine_cc
1         s(1)         age
2         f(2)         gear
3         f(3)         fuel_type
4         f(4)         province
5         f(5)         brand_grouped
6         l(6)         leasing
7         l(7)         air_condition
8         l(8)         power_steering
9         l(9)         power_mirror
10        l(10)        power_window
```

Figure 36: Generalized Additive Model (GAM)

The linear GAM performed well, achieving a Pseudo R^2 of 0.85 and explaining 85% of the variation in vehicle prices. Due to limitations in Python’s GAM implementation, model adequacy checks were conducted in R, with relevant code provided in the Colab file.

s(0) [log_engine_cc], s(1) [age], have EDoF of approximately 14, which confirms that the relationships with price are highly non linear, validating the decision to not consider the OLS.Among categorical features, Brand, Transmission, gear, fuel type, leasing, air condition, power steering were significant price drivers, whereas Province and Power Windows were found to be statistically not significant.

Engine capacity emerged as a primary driver, reflecting market segmentation and possible tax effects, while the age curve captured standard depreciation that reverses for vehicles older than 50 years, indicating entry into the vintage market.

Some equipment features, like air conditioning and power steering, showed counter-intuitive effects. Wide confidence intervals in the partial dependence plots suggest these results are likely due to limited data or bias rather than true market behavior.

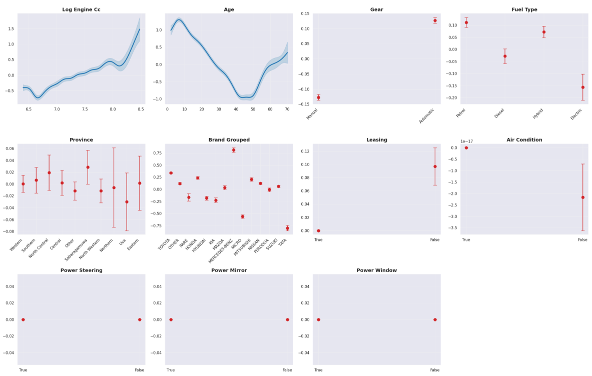


Figure 37: Partial Dependence Plots

Provincial price differences were minor and statistically unstable, indicating that location has a limited impact on online vehicle pricing.

Conclusion

The analysis demonstrates that second-hand vehicle prices in Sri Lanka are shaped by a combination of structural market factors, mechanical attributes, comfort features, and non-linear depreciation dynamics. The dataset is highly heterogeneous, with skewed distributions, extreme values, and dominant brands (notably Toyota and Suzuki), reflecting the real-world diversity of the market.

Brand equity and age remain primary drivers of value: Toyota serves as a mid-market benchmark, retaining value better than budget brands, while luxury and premium vehicles show distinct depreciation patterns. Engine capacity, fuel type, and transmission further segment the market, with hybrid and electric vehicles commanding higher prices, and automatic transmissions consistently priced above manuals. Mechanical features exert non-linear effects, and engine size influences price only within certain segments.

Comfort features, though increasingly standard, continue to add economic value, particularly when bundled, amplifying price appreciation during supply-constrained periods. Geographic location has minimal influence on pricing, reflecting a largely integrated national market, whereas financial accessibility, through leasing, significantly affects median prices, enabling broader market participation.

Modeling with a Generalized Additive Model (GAM) captured these complex non-linear relationships effectively, explaining 85% of price variation while maintaining interpretability. Overall, second-hand vehicle pricing reflects an interplay of brand reputation, depreciation, mechanical performance, feature richness, and market structure, with Toyota, alternative fuels, automatic transmissions, and comfort features consistently sustaining price premiums.

- The complete Google Colab python notebook for this analysis:

Google Colab Python Notebook file